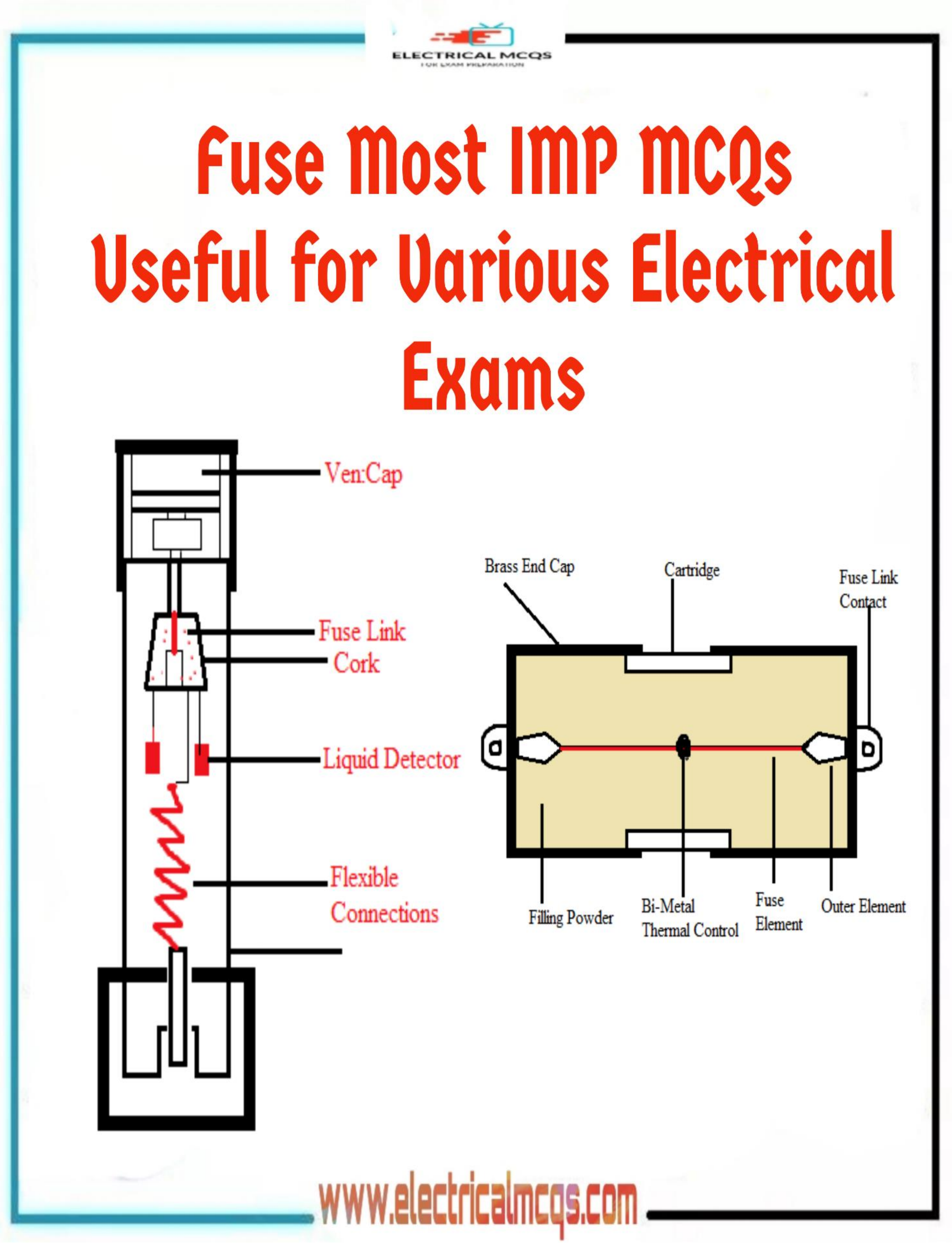
Electrical Fuse MCQ



1. A selection of fuse is based on

a. Steady load
b. Fluctuating load
c. Both (a) & (b)
d. None of these

Answer

Answer. c

2. The rating of fuse wire is always expressed in

a. Volts
b. Amperes
c. Ampere-volt
d. Ampere-hours

Answer

Answer. b

3. Consider the following statements and select the current option.

Statement A: When current more than the fixed value flows in the circuit, fuse wire melts.

Statement B: Fuse wire is made up of a low melting point material.

a. A and B are true and B is the correct explanation of A
b. A and B are true but B is not the correct explanation of A
c. A is true but B is false
d. B is true A is false

Answer

Answer. a

4. HRC fuse provides the best protection against

a. Open circuit

b. Overload

c. Reverse current

d. Short circuit

Answer

Answer. d

5. Which of the following is used in liquid fuses?

a. Transformer oil

b. Sulphur hexafluoride

c. Distilled water

d. Carbon tetrachloride

Answer

Answer. d

6. Main function of the fuse is to

a. Protect the line

b. Open the circuit

c. Protect the appliance

d. Prevent excessive current

Answer

Answer. d

7. For a fuse wire of diameter 'd', fusing current is proportional to

a. $\sqrt{d}$

b. $d^{1.2}$

c. $d^{1.5}$

d. d^3

Answer

Answer. c

8. If a combination of HRC fuse and circuit breaker is used, the circuit breaker operates

a. For low overload currents
b. For short-circuit current
c. Under all abnormal currents
d. Combination is never used in practice

Answer

Answer. a

9. A fuse is inserted in

a. Neutral
b. Earth continuity conductor
c. Phase
d. Both phase and neutral

Answer

Answer. c

10. On what factors does the current carrying capacity of a fuse depend?

I. size of fuse wire
II. fuse material
III. surrounding of the fuse

Choose the correct answer from the options given below.

a. Only I
b. Only I and III
c. Only II and III
d. I, II and III

Answer

Answer. d

11. The function(s) performed by fuse is (are)

i. to carry the normal working current safely without heating
ii. to break the circuit when current exceeds the limiting current choose the correct

Answer from the options given below.

a. Only (i)
b. Only (ii)
c. Both (i) and (ii)
d. Neither (i) nor (ii)

Answer

Answer. c

12. A fuse is never inserted in

a. Neutral wire
b. Negative of DC circuit
c. Positive of DC circuit
d. Phase line

Answer

Answer. a

13. Consider the following statements

HRC fuse as compared to a rewireable fuse has

A. No ageing effect
B. High speed of operation
C. High rupturing capacity

a. Only A and B
b. Only B
c. Only B and C
d. A, B and C

Answer

Answer. d

14. The fusing current depends on

  a. Diameter of fuse wire
  b. Number of wire
  c. Length of the fuse wire
  d. All of the above

Answer

Answer. d

15. The least expensive protection for over-current in a low voltage system is

  a. Rewirable fuse
  b. Isolator
  c. Circuit breaker
  d. Air-break switch

Answer

Answer. a

16. A fuse is provided in an electric circuit to

  a. Reduce the power consumption in the circuit
  b. Limit the current in the circuit
  c. Safeguard the circuit against heavy current
  d. Increase the current in the circuit

Answer

Answer. c

17. HRC fuse has rupturing capacity of capability_______ in comparison with a circuit breaker.

a. Equal
b. Greater than
c. Less than
d. None of these

Answer

Answer. c

18. For better performance the suitable one is

a. Open type fuse
b. Kit-kat fuse
c. HRC fuse
d. Cut-outs

Answer

Answer. c

19. The function of the fuse in an electrical circuit is to protect

a. Insulation resistance
b. Over loading of the circuit
c. Over voltage
d. Insulation resistance and excess voltage

Answer

Answer. b

20. The characteristics of the fusing element should be a wire of

a. Low resistance
b. High resistance
c. High melting point
d. Low resistance & low melting point

Answer

Answer. d

21. Regarding HRC fuses which statement is not correct?

a. They do not deteriorate with time
b. They have inverse time current characteristics
c. Operation of the fuse is very reliable
d. It is not required to be replaced after each operation.

Answer

Answer. d

22. Fusing factor of a fuse is defined as the ratio of

a. Maximum fusing current to the prospective current
b. Maximum fusing current to the cutoff current
c. Minimum fusing current to the rated carrying current
d. Maximum fusing current to the rated carrying current

Answer

Answer. c

23. The rating of fuse wire is always expressed in

a. Ampere-hours
b. kWh
c. Amperes
d. None of the above

Answer

Answer. c

24. According to fuse law, the current carrying capacity varies as

a. diameter
b. $(\text{diameter})^{1.5}$
c. $(\text{diameter})^{0.5}$

d. None of these

Answer

Answer. b

25. Fuse material must have

a. High melting point and high specific resistance
b. Low melting point and high specific resistance
c. Low melting point and low specific resistance
d. High melting and low specific resistance

Answer

Answer. c

26. A fuse is provided in an electric circuit for

a. Safeguarding the installation against heavy current
b. Reducing the current flowing the circuit
c. Reducing the power consumption
d. All options are correct

Answer

Answer. a

27. How should a fuse be installed in a circuit to insure proper operation?

a. Parallel to the load
b. Series with the load
c. In any way possible
d. At the ground point

Answer

Answer. b

28. Which among these fuse is very fast in operation?

a. Semiconductor fuses

b. High rupturing capacity fuse

c. Cartridge type

d. Kit kat type

Answer

Answer. a

29. A good fuse wire must have

a. High melting point

b. High ohmic losses

c. High conductivity

d. High deterioration

Answer

Answer. c

30. Fusing current is the

a. Maximum current at which the fuse element will get heated

b. Rated current of a fuse

c. The minimum current at which the fuse element will get heated

d. The maximum current at which the fuse element will melt

Answer

Answer. d

31. The primary function of a fuse is to

A. Open the circuit

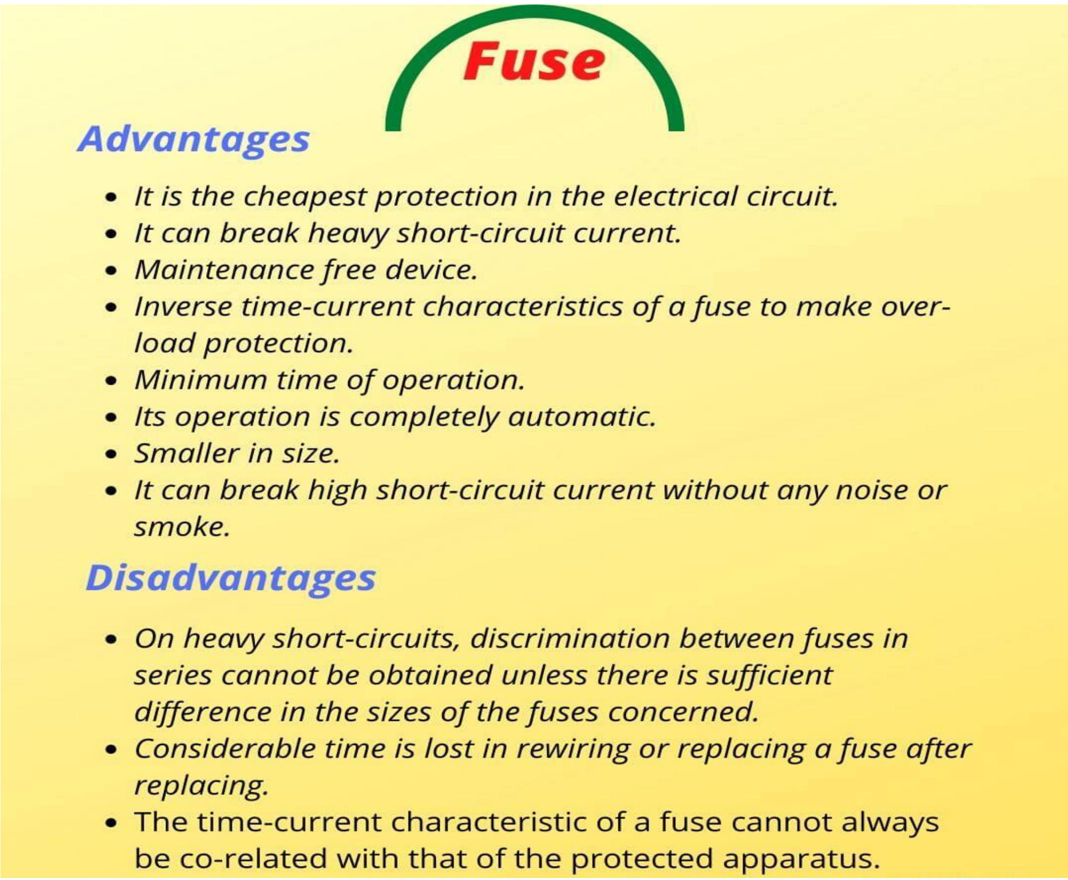
B. Protect the appliance

C. Protect the line

D. Prevent excessive currents from flow through the circuit

Answer

D. Prevent excessive currents from flow through the circuit



32. The fuse rating is expressed in terms of

A. Current

B. Voltage

C. VAR

D. KVA

Answer

A. Current

33. if use is never inserted in

A. Neutral wire

B. Negative of DC circuit

C. Positive of DC circuit

D. Phase line

Answer

A. Neutral wire

34. Fuses have got advantages of

A. Cheapest type of protection

B. Inverse time current characteristic

C. No Maintenance

D. Current limiting effect under short-circuit conditions

E. All of the above

Answer

E. All of the above

35. Protection by fuses is generally not used beyond

A. 20 A

B. 50 A

C. 100 A

E. 200 A

Answer

C. 100 A

36. The fuse wire in DC circuit is inserted in

A. Negative circuit only

B. Positive circuit only

C. Both A and B

D. Either A or B

Answer

C. Both A and B

37. On which off the following effects of electric current a fuse operates?

A. Photoelectric effect

B. Electrostatic effect

C. Heating effect

D. Magnetic effect

Answer

C. Heating effect

38. A fuse in a motor circuit provides protection against

A. Overload

B. Short-circuit and overload

C. Open circuit, short-circuit and overload

D. None of the above

Answer

B. Short-circuit and overload

39. Semi-closed rewireable fuses have....... breaking capacity

A. Low

B. High

C. Medium

D. Extremely high

Answer

A. Low

40. In HRC fuse the time between cut-off and final current zero is called the

A. Pre-arcing time

B. Arcing time

C. Total operating time

D. None of the above

Answer

B. Arcing time

High-Rupturing Capacity (H.R.C) Cartridge Fuse

Advantages

- *It is very simple and easy to install.*
- *It is a maintenance-free fuse.*
- *High-speed operation.*
- *It provides reliable discrimination.*
- *It is capable to clear high as well as low fault current.*
- *It is cheaper than other same category fuses.*
- *It gives you reliable performance for a long time.*
- *For inverse time-current characteristics, it is suitable for over-load protection.*
- *The Operation is very quick.*

Disadvantages

- *This has to be replaced after each operation.*
- *In every operation, it produces heat which affects associate contacts.*
- *Interlocking is not possible.*

41. HRC fuse as compared to a rewireable fuse has

A. No ageing effect

B. High speed of operation

C. High rupturing capacity

D. All of the above

Answer

D. All of the above

42. If a combination of HRC fuse and a circuit breaker is employed, the circuit breaker operates for

A. Short circuit current

B. Low overload currents

C. Under all abnormal currents

D. All of the above

Answer

B. Low overload currents

43. By which of the following methods major portion of the heat generated in a HRC fuse is dissipated?

A. Radiation

B. Convection

C. Conduction

D. None of the above

Answer

C. Conduction

44. HRC fuses provide best protection against

A. Overload

B. Reverse current

C. Open circuits

D. Short-circuits

Answer

D. Short circuits

45. The delay fuses are used for the protection of

A. Motors

B. Power outlet circuits

C. Fluorescent lamps

D. Light circuits

Answer

A. Motors

46. Which of the following is used in liquid fuses?

A. Transformer oil

B. Sulphur hexafluoride

C. Distilled water

D. Carbon tetrachloride

Answer

D. Carbon tetrachloride

47. A fuse performs

A. Only detection

B. Only interruption

C. Both A and B

D. None of the above

Answer

C. Both A and B

48. A fuse element should have......... melting point

A. High

B. Low

C. Same

D. None of the above

E. All of the above

Answer

B. Low

49. A fuse melts well........ the first peak of fault current is reached

A. Before

B. After

C. Same time

D. None of the above

Answer

A. Before

50. The contact resistance of a manually operated switch is

A. Zero

B. Very high

C. Very low

D. None of the above

Answer

C. Very low

51. Which one of these is not a manually operated switch?

A. Thumbwheel switch

B. Rotary selector switch

C. Crossbar switch

D. Toggle switch

Answer

C. Crossbar switch

52. A switch should have

A. High insulation resistance

B. Low insulation resistance

C. Insulation resistance equal to contact resistance

D. None of the above

Answer

A. High insulation resistance

53. A Thumbwheel switch

A. Has an operating wheel which has numbers on it

B. Is an alternative form of rotary switch

C. Requires half the operating torque required in a rotary switch

D. Supports the features A, B and C

Answer

D. Supports the features A, B and C

54. When a membrane key switch is pressed

A. The row conductor sheet gets separated from the column contact conductor sheet

B. The row conductor sheet comes into contact with column conductor sheet

C. A plunger exerts a pressure on the carbon button and it touches the contact print on the circuit board

D. None of the above

Answer

B. The row conductor sheet comes into contact with column conductor sheet

55. According to the fuse law, the current carrying capacity varies as

A. Diameter

B. $\text{Diameter}^{1.5}$

C. $\text{Diameter}^{1/2}$

D. 1/Diameter

Answer

B. $\text{Diameter}^{1.5}$

56. The fuse blows off by

A. Arcing

B. Burning

C. Melting

D. None of the above

Answer

C. Melting

57. The material bast suited for manufacturing of fuse wire is

A. Silver

B. Copper

C. Aluminium

D. Zinc

Answer

A. Silver

58. A fuse is provided in an electric circuit for

A. Safeguarding the installation against heavy current

B. Reducing the current flowing in the circuit

C. Reducing the power consumption

D. All of the above

Answer

A. Safeguarding the installation against heavy current

59. A fuse is normally a

A. Power limiting device

B. Voltage limiting device

C. Current limiting device

D. Power factor correcting device

Answer

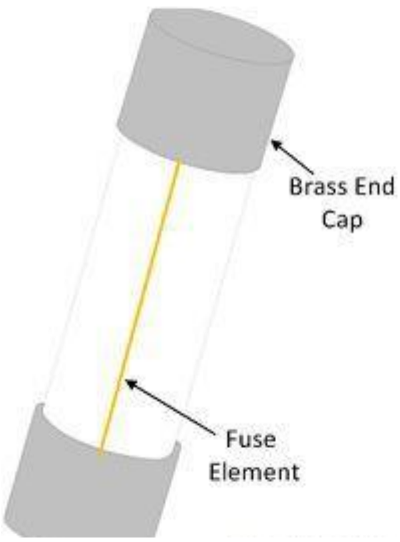
B. Current limiting device

Important Notes About Fuse

What is Fuse?

The fuse is a device used in an electrical circuit for protecting electrical devices against overloads and short circuit. It is the simplest and cheapest devices used for interrupting an electrical circuit under short circuit or excessive overload current magnitudes.

It is used for overload or short circuit protection in high voltage up to 66 KV and low voltage up to 400 kV. In some places, their use is restricted to those applications where their performance characteristics are especially suitable for the current interruption.



The working of the fuse depends on the heating effect of the current. In normal operating condition, the normal current passes through the fuse. The heat develops in the fuse element because of the normal current and this heat is dissipated by the help of the surrounding air. Thus, the temperature of the fuse is kept below the melting point.

When the fault occurs the short circuit current passes through the fuse element. The magnitude of the current is much high as compared to the normal current. The short circuit current generates excessive heat in the fuse element. Thereby, the element becomes melts and break. The fuse protects the machine or apparatus from short circuit or overload current.

The fuse element is made of the highly selected metal conductor. The cartridge holds the fuse element. The main function of the fuse of the element is to allow the normal current to flows through the fuse and break the circuit when the high magnitude current passes through it.

Advantages of an Electrical Fuse

- It is the cheapest form of protection, and it does need any maintenance.
- Its operation is completely automatic and requires less time as compared to circuit breakers.
- The smaller sizes of fuse element impose a current limiting effect under short-circuit conditions.
- Its inverse time-current characteristic enables its use for overload protection.

Disadvantages of an Electrical Fuse

- Considerable time is required in replacing a fuse after the operation.
- The current-time characteristic of a fuse cannot always be correlated with that of the protective device.

Fuses are used for the protection of the cables in low voltages light, and power circuits and for transformers of rating not exceeding 200 KVA, in the primary distribution system. Fuses are used in low and moderate voltage applications where the frequent operation is not expected or where the use of a circuit breaker is uneconomical.

Fuse Wire Materials

The material used for fuse elements must be of low melting point, low ohmic loss, high conductivity (or low resistivity), low cost and free from detraction. The material used for making fuse element has a low melting point such as tin, lead, or zinc. A low melting point is, however, available with a high specific resistance metal shown in the table below.

Metal	Melting Point in Celsius	Specific Resistance	Value of Fuse constant k for d in mm
Silver	980	16	-
Tin	240	112	12.8
Zinc	419	60	-
Lead	328	210	10.8
Copper	1090	17	80
Aluminium	665	28	59

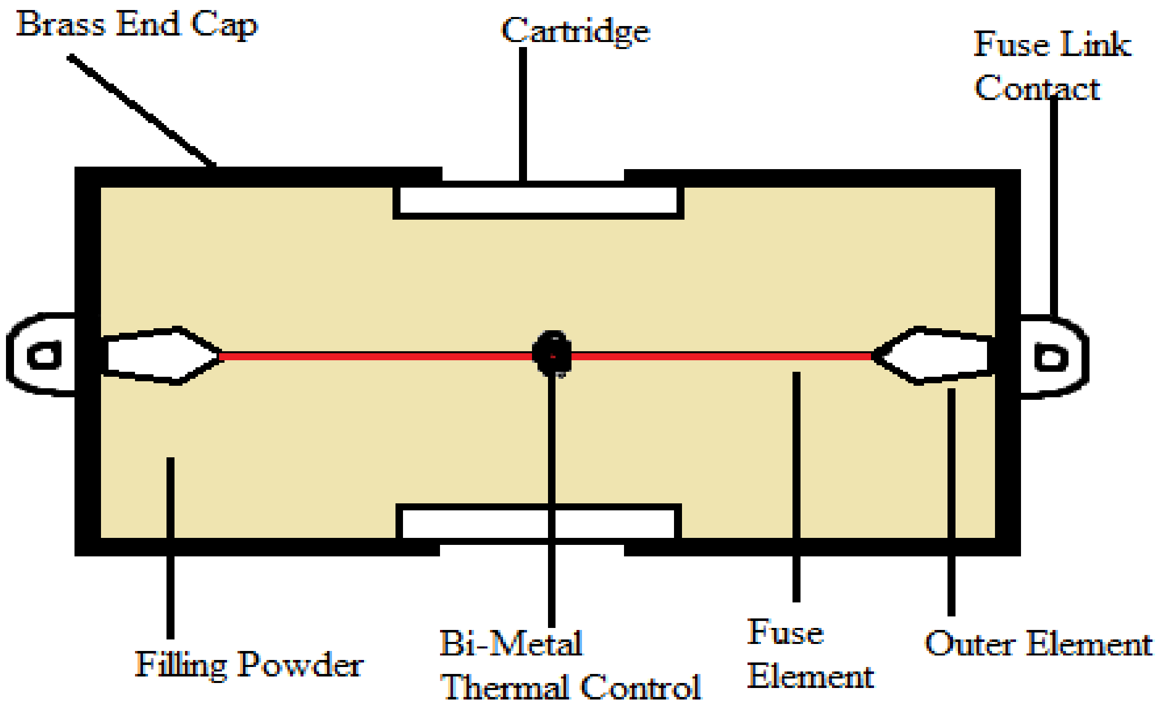
The material mainly used for fuse element are tin, lead, silver, copper, zinc, aluminium, and an alloy of lead and tin. An alloy of lead and tin is used for small current rating fuses. For current exceeding 15A this alloy is not used as the diameters of the wire will be larger and after fusing the metal released will be excessive.

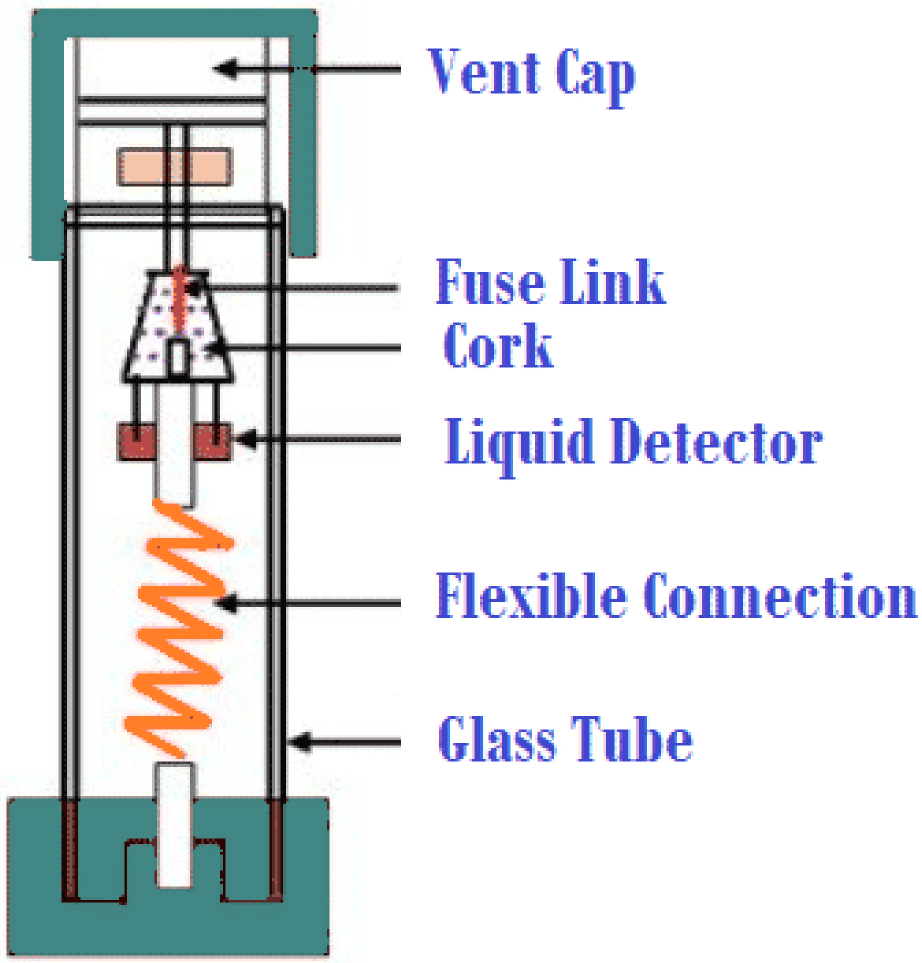
Beyond 15A rating circuit, copper wire fuses are employed. Copper wire suffers from the drawbacks that it operates at a rather high temperature if a reasonably low fusing factor is desired. There is thus a tendency for the wire to overheat with the result that its cross-sectional area and fusing current are gradually reduced, and premature melting of the wire may occur.

Silver is used as a fuse element because it has the following advantages

- It does not get oxidized, and its oxide is unstable.
- The conductivity of silver does not deteriorate with oxidation.
- Owing to its high conductivity the mass of molten metal to be handled is minimized and thus operating speed is fast.

But silver is very costly as compared to other metals thereby copper or alloy of lead-tin is mostly used as a fuse wire. Zinc in strip form only is also used as a fuse element because it does not melt very quickly with a small overload.


Brass End Cap
Cartridge
Fuse Link
Contact
Filling Powder
Bi-Metal
Thermal Control
Fuse
Element
Outer Element


Vent Cap
Fuse Link
Cork
Liquid Detector
Flexible Connection
Glass Tube